CHEMISTRY ONLY

27 CARBOXYLIC ACIDS

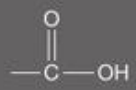
Previously you have seen how alcohols can be oxidised to form a class of organic compounds called carboxylic acids. Here we look at their structures and some of their reactions in more detail.



▲ Figure 27.1 Stinging nettles inject methanoic acid.

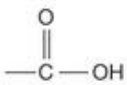
LEARNING OBJECTIVES

- Know that carboxylic acids contain the functional group



- Understand how to draw structural and displayed formulae for unbranched-chain carboxylic acids with up to four carbon atoms in the molecule, and name each compound
- Describe the reactions of aqueous solutions of carboxylic acids with metals and metal carbonates
- Know that vinegar is an aqueous solution containing ethanoic acid

Acids such as ethanoic acid are known as **carboxylic acids** and they all contain the functional group **–COOH**



The most familiar one is ethanoic acid (old name *acetic acid*). **Vinegar** is *an aqueous solution containing ethanoic acid*.

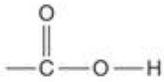
Carboxylic acids are formed by oxidation of the corresponding alcohols, for example ethanoic acid is formed when ethanol is left in the air or oxidised using potassium dichromate(VI) in dilute sulfuric acid. If you need a reminder, have at look at Chapter 26, pages 289–290.

DRAWING AND NAMING THE CARBOXYLIC ACIDS

Again, we are only going to be looking at the four simplest carboxylic acids: methanoic acid, ethanoic acid, propanoic acid and butanoic acid. These are shown in Table 27.1.

HINT

If you are asked for a displayed formula in an exam, you should draw all the bonding in the COOH group as shown here. If you just want to write the formula in an equation, it is normal to write it as, for example, CH_3COOH.

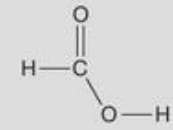
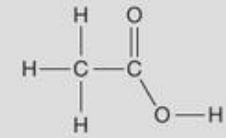
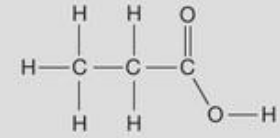
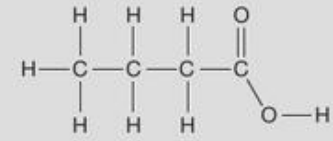


▲ Figure 27.2 The carboxylic acid functional group

KEY POINT

It is really important to notice that when we name a carboxylic acid we count the total number of carbon atoms, including the one in the –COOH group. So propanoic acid has three carbon atoms in total, which is why it starts with 'prop'. The –COOH group is always on the end carbon, so there is no need to insert a number in the name to represent the position of the functional group.

Table 27.1 Formulae and names of the first four members of the carboxylic acid homologous series

Name	Molecular formula	Structural formula	Displayed formula
methanoic acid	CH_2O_2	HCOOH	
ethanoic acid	$C_2H_4O_2$	CH_3COOH	
propanoic acid	$C_3H_6O_2$	CH_3CH_2COOH	
butanoic acid	$C_4H_8O_2$	$CH_3CH_2CH_2COOH$	

ACID PROPERTIES OF THE CARBOXYLIC ACIDS

Ethanoic acid and the other carboxylic acids are weak acids with a pH of about 3–5, depending on the concentration of the solution. They will turn blue litmus paper red, and react with all the things you expect acids to react with.

In the reactions described here, we are taking ethanoic acid as an example because it is the one you are most likely to use. All the others will behave similarly, and if necessary all you need to do is change the formula from CH_3COOH to, for example, CH_3CH_2COOH.

REACTION WITH METALS



▲ Figure 27.3 Reaction between magnesium ribbon and ethanoic acid.

Dilute ethanoic acid reacts with metals in the same way as other dilute acids such as hydrochloric acid or sulfuric acid, only *more slowly*. Remember *metals react with acids to give a salt and hydrogen* (see Chapter 17, page 174).

For example, dilute ethanoic acid reacts with magnesium with a lot of fizzing to produce a salt and hydrogen. The salt formed is magnesium ethanoate, which is soluble in water and so you get a colourless solution:

$$Mg(s) + 2CH_3COOH(aq) \rightarrow (CH_3COO)_2Mg(aq) + H_2(g)$$

Students frequently have problems with the formulae of salts like magnesium ethanoate. You have to remember that this is an ionic compound, and so